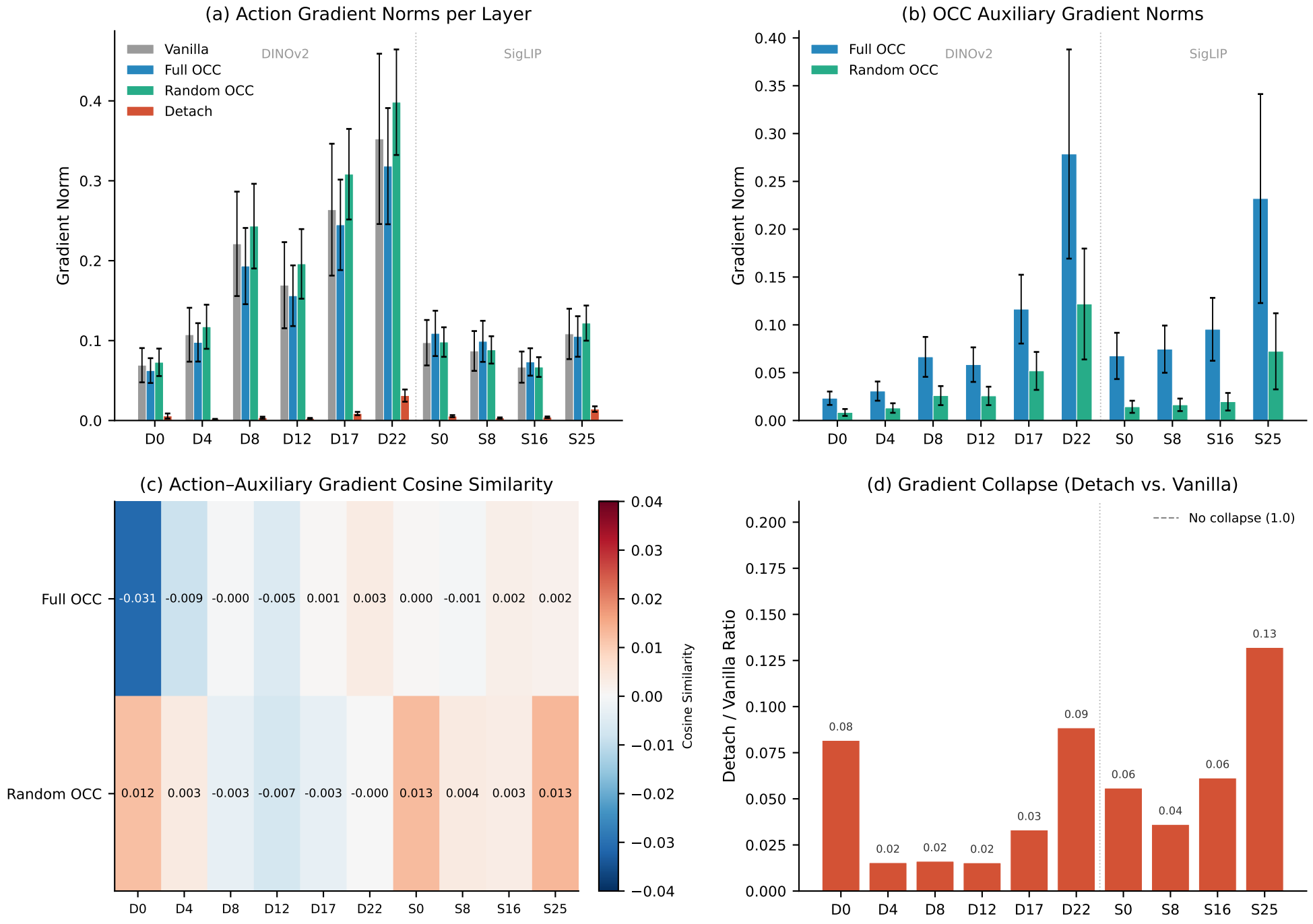
(a) Action Gradient Norms per Layer
Vanilla
Full OCC
Random OCC
Detach
DINOv2
SigLIP
Gradient Norm
0.0
0.1
0.2
0.3
0.4
D0
D4
D8
D12
D17
D22
S0
S8
S16
S25
(b) OCC Auxiliary Gradient Norms
Full OCC
Random OCC
DINOv2
SigLIP
Gradient Norm
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
D0
D4
D8
D12
D17
D22
S0
S8
S16
S25
(c) Action–Auxiliary Gradient Cosine Similarity
Full OCC
Random OCC
-0.031
-0.009
-0.000
-0.005
0.001
0.003
0.000
-0.001
0.002
0.002
0.012
0.003
-0.003
-0.007
-0.003
-0.000
0.013
0.004
0.003
0.013
D0
D4
D8
D12
D17
D22
S0
S8
S16
S25
Cosine Similarity
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
(d) Gradient Collapse (Detach vs. Vanilla)
No collapse (1.0)
Detach / Vanilla Ratio
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.08
0.02
0.02
0.02
0.03
0.09
0.06
0.04
0.06
0.13
D0
D4
D8
D12
D17
D22
S0
S8
S16
S25